



GROUP TECHNICAL SOLUTIONS (GTS)

TECHNICAL REPORT FOR PETRONAS CHEMICAL LOW DENSITY POLYETHYLENE SDN BHD (PCLDPE)

APC Functional Design Specifications (FDS)

January 2020

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1 EXECUTIVE SUMMARY

The purpose of this document is to provide the APC engineering design specifications and its control strategy for the APC application which was successfully implemented in PCLDPE. This document is an updated document based on the review and comments of the APC functional design specifications throughout the project implementation and the concluded applications as installed, tested and commissioned.

Advanced Process Control application was implemented in PCLDPE in order to achieve several objectives as below:

- Minimize variance in MFI and Gloss during steady state operation
- Minimize transition times by automating transitions
- Maximizing production rate without compromising quality

There are total of 6 MVs, 4 CVs and 17 FFs identified during model identification. Generally, all APC variables are located around reactor unit in order to stabilize and optimize the operation at the heart of the plant. Another application, Grade Transition Manager, is modified according to recent revamp on the APC in order to minimize the transition time and to automate the transition between grades.

This document shall provide the details for all the controllers which cover process description, constraints, operational objectives, APC controller design and objectives, list of all the variables (manipulated/controlled/feedforward) used in APC, APC model / gain matrices, APC controller tuning and internal calculation settings.

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2 INTRODUCTION

The purpose of this document is to provide the APC engineering design specifications and its control strategy for the APC application which was successfully implemented in PCLDPE. This document is an updated document based on the review and comments of the APC functional design specifications throughout the project implementation and the concluded applications as installed, tested and commissioned.

3 APC DESIGN BASIS SUMMARY

4.1 General APC Control Objectives

General control objectives of the Advanced Process Control (APC) applications are designed to reduce variability of the key qualities and to maintain them to their constraints as close as possible. This is achieved through model predictive control, feed forward disturbance rejection and overall / unit operation optimization. Thus, APC applications are designed to meet the following specific objectives:

- Maintain a stable process for the plant / smoother operation.
- Maximize plant throughput and conversion while yielding more products.
- Optimize utility consumption such as steam, fuel and electricity.
- Stabilize product quality and optimize closer to the required specifications.
- Maintain overall targets to their constraints as close as possible.
- Minimize losses (e.g. product loss, recycle components).
- Improve plant process safety.

4.2 APC Control Design and Controller Objectives

Advanced Process Control application was implemented in PCLDPE in order to achieve several objectives as below:

- Minimize variance in MFI and Gloss during steady state operation
- Minimize transition times by automating transitions
- Maximizing production rate without compromising quality

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These objectives can be achieved by having few APC components which are:

- a) Polymer Inferential Properties (Aspen IQ)
- b) Non-Linear Advanced Process Control
- c) Process Recipe and Transition Manager (Grade Transition Manager)

Based on components above, generally, there is only 1 nonlinear controller "PETLIN_ASV2020" used in PCLDPE. This controller was designed to control MFI, Gloss during steady state and transition and production rate according to the unit constraints.

After implementation of APC and all the components, the improvement should be reflected with reduction in quality variance, less batch downgrading, and less transition substandard production.

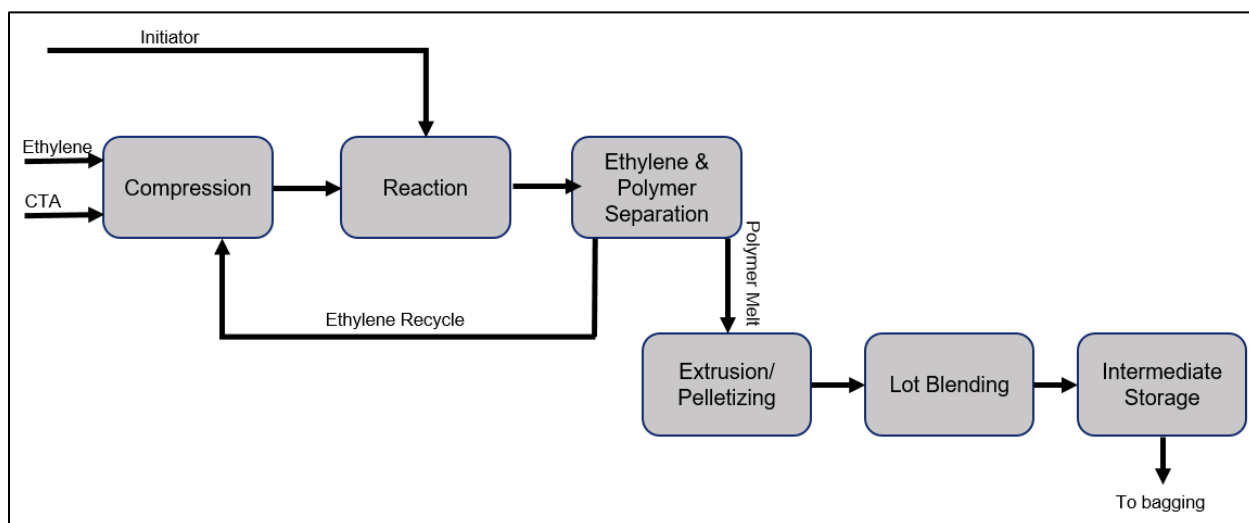
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4 APC CONTROLLER DESIGN

4.1 Process Description/ Constraints/Objectives

PCLDPE produces 290 000 MTPA (after revamp project in 2016) of Low Density Polyethylene (LDPE) by using Stamicarbon Clean Tubular Reactor (CTR) technology.

Ethylene feed and Chain Transfer Agent (CTA), Propylene/Propane are from PC Olefin, Glycol and Derivatives (PCOGD). The polymerization of ethylene is carried out in a tubular reactor at a very high pressure (2650 barg) in the presence of peroxide as initiator and propylene/propane as the chain transfer agent. The reactor pressure is controlled by 12PIC0103 and is not changed during normal operation. Generally, higher reactor pressure would increase the polymerization velocity shifting the temperature peak to the front, and decreasing initiator consumption. The Peroxide enables the polymerization on the reactor to proceed at low temperatures of 160 to 300 °C. The reactor temperature, initiator type and CTA flow are manipulated to produce the required LDPE grade.



Ethylene is compressed to the reactor pressure of 2650 bar in the following steps:

- 1) Booster compressor - Pressurizes the low pressure recycle gas from the low pressure separator
- 2) Primary compressor - Pressurizes gas from booster compressor, Ethylene and CTA from battery limit to secondary compressor at 250 bar.
- 3) Secondary compressor - Pressurizes the gas to the reactor pressure at 2650 bar.

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The compressed ethylene is then fed to the tubular reactor. The reactor consists of a Preheater and 4 Polymerization sections (4 initiator dosing points). The preheating is a 2-stage process which start with reactor hot water (return stream from the reactor) followed by high pressure steam in the 2nd stage. The ethylene from the preheater at temperature of 165 – 170 °C is fed to the 1st polymerization section where polymerization is started with the addition of initiator. The initiator is prepared in an initiator vessel from where they are pumped to the reactor.

The addition of initiator result in rapid temperature rise to 260 to 300 °C in the reactor. The reactor tubes are cooled externally to remove heat of reaction. The cooling is achieved using hot water. When the temperature of the reaction mixture has dropped to an economically low level, the 2nd initiator dosing is done. This again result in temperature rise and cooling thereafter. The 3rd and 4th initiator dosing is carried out similarly to obtain optimum conversion in the reactor.

The product quality and conversion per pass is dependent on the temperature profile in the reactor, type and amount of initiator and type and amount of CTA. The reactor temperature and pressure should be within the specified range to prevent phase separation and decomposition of ethylene. The temperature profile is maintained by changing the quantity of initiator, the temperature and flow of reactor hot water. In normal operations, the quantity of initiator changes automatically based on the highest temperature in the region to maintain the reactor temperature profile. As of now, the reactor temperature profile provides the best indication of the extent of polymerization (conversion) in the reactor.

Each of the 4 sections has a temperature controller (called Peak temperature controller). The temperature is controlled by changing the flow of initiator to the reactor. The initiator flow in turn in controlled by altering the stroke frequency of individual initiator pump. The peak temperature controller comprises a high selector which finds the maximum temperature out of the sensors in the region, this temperature is then used as input to the peak temperature controller. The table below summarizes the sensors, selector switches and controllers used for individual peak temperature control.

Peak No	Sensors	Output of high selector	Peak temperature controller
1	12TE0202A – 12TE0219A	12TY0710	12TIC0720/12TIC0730
2	12TE0226A – 12TE0240A	12TY0810	12TIC0820/12TIC0830
3	12TE0302A – 12TE0312A	12TY0910	12TIC0920/12TIC0930
4	12TE0320A – 12TE0325A	12TY0940	12TIC0930/12TIC0950

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There is possibility that 4th peak shows a lower temperature than the actual due to layer of polyethylene deposit. So, the 4th peak has a gain scheduler operating in conjunction with override control 12TIC0333. The gain scheduler changes the proportional gain of controller with respect to changes of the process value from the set point. For the override controller, TIC0333, a separate thermocouple is provided for measurement. The set point of TIC0333 is kept at a low value, if the TE0333 temperature rises above this value, the override controller become active. A low selector then compares the output of active 4th peak with the TIC0333 output and the lower of the two is the feed to the 4th initiator pump

The polymer and unconverted ethylene from the reactor are expanded through a pressure control valve, cooled and separated in 2 stages which are high pressure separator and low pressure separator. The unconverted ethylene from both separators is recycled back through compressor to the reactor. A portion of the gas from the high pressure separator is removed as purge to maintain the level of inert and impurities which can accumulate in the process.

The polyethylene from the low pressure separator flows by gravity to the extruder. The extruder is provided with an underwater pelletizer. The pellets from the pelletizer are dried in a centrifugal drier. The dried pellet are pneumatically conveyed through a product hopper to lot composition and storage area.

4.2 APC Controller Objectives

Main objectives of PETLIN_ASV2020 are as below:

In grade objectives:

- Minimize variance in MFI and Gloss during steady state operation.
- Maximizing production rate without compromising quality subject to constraint

Transition objectives:

- Minimize transition time and loss of production and product downgrading due to transition

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4.2.1 PETLIN_ASV2020 controller

Table 1 : MV/CV Control Objectives for Melt Flow Index (MFI)

Manipulated Variables (MV)	Controlled Variables (CV)	APC Controller Objectives
DIEPRESS.SV (Dieplate Pressure)	MFI (Melt Flow Index Analyser)	To control MFI at target (i.e. 5) as set by the operator or APR/TM. Main handle is die plate pressure. However, this is not a real controller. DIEPRESS.SV will be used as external target for DIEPRESS.PV. MFI is predicted value from inferential model. Mainly used all the time. Bias update is configured to receive input from 2 sources which are analyser (13AI0101) and lab measurement.

Table 2: MV/CV Control Objectives for Die plate Pressure

Manipulated Variables (MV)	Controlled Variables (CV)	APC Controller Objectives
APCTA (Propylene Flow)	DIEPRESS.PV (Die plate Pressure)	To control die plate pressure at its target and its target coming from DIEPRESS.SV (sent by APC)

Table 3: MV/CV Control Objectives for Gloss

Manipulated Variables (MV)	Controlled Variables (CV)	APC Controller Objectives
APTEMP1 (Peak Temp Zone 1) APTEMP2 (Peak Temp Zone 2) APTEMP3 (Peak Temp Zone 3) APTEMP4 (Peak Temp Zone 4) 11FIC0401 (Purge gas flow)	GLOSS (Gloss)	To control Gloss within limit as set by the operator or APR/TM. Main handle is Peak temperature. GLOSS predicted value from inferential model. It is not used during high MFI production. Purge gas will be set to be minimized. Minimum purge is favourable.

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Table 4: MV/CV Control Objectives for Prodspex

Manipulated Variables (MV)	Controlled Variables (CV)	APC Controller Objectives
APTEMP1 <i>(Peak Temp Zone 1)</i> APTEMP2 <i>(Peak Temp Zone 2)</i> APTEMP3 <i>(Peak Temp Zone 3)</i> APTEMP4 <i>(Peak Temp Zone 4)</i> 11FIC0401.PV <i>(Purge gas flow)</i>	PRODSPEX <i>(Production Rate)</i>	To maximize production rate to its high limit as set by the operator or APR/TM. Main handle is Peak temperature. PRODSPEX is used all the time to optimize the production rate. Purge gas will be set to be minimized to maximize Prodspex.

Table 2: FF/CV Control Objectives for MFI

Feed Forward (FF)	Controlled Variables (CV)	APC Controller Objectives
11FIC0602.PV <i>(CTA Flow)</i> 13JI0101.PV <i>(Extruder Torque)</i> Flag_Lo <i>(Flag for Low MFI grade product)</i> Flag_Mid <i>(Flag for Mid MFI grade product)</i> Flag_High <i>(Flag for High MFI grade product)</i>	MFI <i>(Melt Flow Index)</i>	To capture / reject the disturbance created from all disturbances. Low MFI grade refer to N103X, N109X. Mid MFI grade refer to N125Y, C150Y, C170X. High MFI grade refer to M022X, M050X. Flags are used to let the controller know what grade is currently produced and to use dedicated model for that grade. Flags is configured at GTM.

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Table 6: FF/CV Control Objectives for Die plate pressure

Feed Forward (FF)	Controlled Variables (CV)	APC Controller Objectives
12PIC0103SV <i>(Reactor Pressure)</i> 12TIC0401PV <i>(RHW Temp PV)</i> RXSTARTTEMP <i>(Rxn Start Temp)</i> APTEMP1PV <i>(Peak Temp Zone 1)</i> APTEMP2PV <i>(Peak Temp Zone 2)</i> APTEMP3PV <i>(Peak Temp Zone 3)</i> APTEMP4PV <i>(Peak Temp Zone 4)</i> 11FIC0401.PV <i>(Purge gas flow)</i> CTARATIO <i>(CTAFLOW/PRODSPEX)</i> 13TI0116B.PV <i>(Extruder temperature)</i>	13PI0107B.PV <i>(Die plate Pressure)</i>	To capture / reject the disturbance created from all disturbances.

Table 3: FF/CV Control Objectives for Gloss

Feed Forward (FF)	Controlled Variables (CV)	APC Controller Objectives
12PIC0103SV <i>(Reactor Pressure)</i> 11FIC0602.PV <i>(CTA Flow)</i> RXSTARTTEMP <i>(Rxn Start Temp)</i>	GLOSS <i>(Gloss Lab)</i>	To capture / reject the disturbance created from reactor pressure, CTA Flow and Reaction start temperature.

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Table 4: FF/CV Control Objectives for Prodspex

Feed Forward(FF)	Controlled Variables (CV)	APC Controller Objectives
11FIC0602.PV (CTA Flow) 11TI0302.PV (Suction temperature) RXSTARTTEMP (Rxn Start Temp) DTALLZONE (Summation of all DIFFTEMP)	PRODSPEXPV (Production Rate)	To capture / reject the disturbance created from all disturbances. DIFFTEMP1 (Peak 1 – 11TIC0102) DIFFTEMP2 (Peak 2 – 12TI0225A) DIFFTEMP3 (Peak 3 – 12TI0301A) DIFFTEMP4 (Peak 4 – 12TI0319A)

4.3 List of MVs, CVs and FFs

A total of 6 MVs was identified for PETLIN controller as below table

No	Model Tag	DCS Point	Description
1	APCTA.SV	APCTA.RSV	Propylene Flow
2	APTEMP1.SV	APTEMP1.RSV	Peak Temp Zone 1
3	APTEMP2.SV	APTEMP2.RSV	Peak Temp Zone 2
4	APTEMP3.SV	APTEMP3.RSV	Peak Temp Zone 3
5	APTEMP4.SV	APTEMP4.RSV	Peak Temp Zone 4
6	APPURGE.SV	APPURGE.RSV	Purge Gas Flow

A total of 4 CVs was identified for PETLIN controller as below table:

No	Model Tag	DCS Point / IP21 Point	Description
1	MFI	IQ_MFIDIE_ASV20_BA	Melt Flow Index
2	DIEPRESS.PV	13PI0107B_PVFLT.PV	Die Plate Pressure
3	GLOSS	IQ_GLOSSASV20_BA	Gloss Lab
4	PRODSPEX	PRODSPEX_PVFLT.PV	Production Rate

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A total of 17 FFs was identified for PETLIN controller as below table:

No	Model Tag	DCS Point	Description
1	12PIC0103SV	12PIC0103_PVFLT.PV	Reactor Pressure
2	12TIC0401PV	12TIC0401_PVFLT.PV	RHW Temp PV
3	RXSTARTTEMP	12TIC0102A_PVFLT.PV	Rxn Start Temp (Preheater)
4	APTEMP1PV	APTEMP1.RSV	Peak Temp Zone 1
5	APTEMP2PV	APTEMP2.RSV	Peak Temp Zone 2
6	APTEMP3PV	APTEMP3.RSV	Peak Temp Zone 3
7	APTEMP4PV	APTEMP4.RSV	Peak Temp Zone 4
8	11FIC0401.PV	APPURGE.RSV	Purge Gas Flow
9	CTARATIO	CTARATIO_PVFLT.PV	CTAFLOW/Ethylene Flow
10	13TI0116B.PV	13TI0116B_PVFLT.PV	Extruder Temperature
11	13JI0101.PV	13JI0101_PVFLT.PV	Extruder Torque
12	11TI0302.PV	11TI0302_PVFLT.PV	Suction temperature
13	DTALLZONE	DTALLZONE_PVFLT.PV	Summation of all DIFFTEMP
14	11FIC0602.SV	APCTA.RSV	CTA flow
15	FLAG_LO	Programmed in GTM	Flag for low MFI grade
16	FLAG_MID	Programmed in GTM	Flag for mid MFI grade
17	FLAG_HI	Programmed in GTM	Flag for high MFI grade

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4.4 APC Model and Gain Matrix

Below is the Expected Gain Matrix for APPETLIN controller. This matrix is to indicate the relationship that exist between MV/FF and CV. (+) indicate positive relationship, which when MV is increased, its CV will also increase and vice versa.

CV	Variables	PRODSPEX.PV	MFI	DIEPRESS.PV	GLOSS
MV	APCTA.SV			-	
	APTEMP1.SV	+			-
	APTEMP2.SV	+			-
	APTEM3.SV	+			-
	APTEMP4.SV	+			-
	DIEPRESS.SV		-		
	APPURGE.SV	-			+
DV	11FIC0401.SV			+	
	12PIC0103.PV			+	+
	12TIC0102A.PV	-		-	+
	12TIC0401.PV			-	
	APTEMP1.SV			-	
	APTEMP2.SV			-	
	APTEM3.SV			-	
	APTEMP4.SV			-	
	DTALLZONE	+			
	11TI0302.PV	-			
	CTARATIO			-	
	11FIC0602.PV	+	+		+
	13JI0101.PV		-		
	13TI0116B.PV			-	
	FLAG_HI		X		
	FLAG_MID		X		
	FLAG_LO		X		

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Output Input	PRODSPEX.PV	MFI	DIEPRESS.PV	GLOSS
▶ APCTA.SV	☐		☒	☐
APTTEMP1.SV	☒		☐	☒
APTTEMP2.SV	☒		☐	☒
APTTEMP3.SV	☒		☐	☒
APTTEMP4.SV	☒		☐	☒
DIEPRESS.SV		☒		
11FIC0401.SV	☒		☐	☒
12PIC0103.PV			DV	DV
12TIC0102A.PV	DV		DV	DV
12TIC0401.PV			DV	
DTALLZONE	DV			
11TI0302.PV	DV			
CTARATIO			DV	
11FIC0602.PV		DV		
EXTORQUE		DV		
13TI0116B.PV			DV	
FLAG_LO		DV		
FLAG_MID		DV		
FLAG_HI		DV		

4.5 APC Internal logic Calculation

There are few logics has been formulated in the calculation section. Basically, these logics are made in order to ensure smooth APC operation once commissioned. Below are some examples of logics configured:

- 1) APTTEMP4 MODE: This logic is to switch off MV APTTEMP4 whenever peak temperature 4 is taken over by the override controller TIC0333A

```

IF MASTER_ON_REQ = 1 AND MASTER_ON_STAT = 1 THEN

IF TYA0950BPV =1 OR TYA0930BPV =1 THEN
    SERV_REQ = 0
END IF

END IF

```

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- 2) APCTA_PVTRACK: This logic is to ensure, whenever APC is OFF, the measurement will follow the current value. Thus, there is no bump whenever APC is turned ON. The same logic is duplicated for other MVs (APTEMP1/2/3/4_PVTRACK, DIEPRESS_PVTRACK, APPURGE_PVTRACK) and CV target (MFI_TARGET_TRACK, DIE_TARGET_TRACK)

```

IF MASTER_ON_REQ = 0 or MASTER_ON_STAT = 0 or SRVREQ=0 THEN
MEAS=PVFLT
ELSE
MEAS=PV
END IF

```

- 3) APTEMP1_AWC: This logic is to prevent huge difference between current value to the setpoint given by APC. When the difference greater than the factor configured, APC will restrict the MV setpoint from increase or decreasing by changing the status to be at AWS. The same logic has been configured for other MVs. The factor can be changed by APC engineer through PCWS and it is in engineering unit.

```

IF ABS(RSV-PVFLT)>DIFFTEMP THEN
AWC=3
END IF

```

4.6 APC Controller Tuning

Refer to Appendix 1.

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5 CONCLUSION AND RECOMMENDATION

The APC application also has been commissioned and tested on several product grade such as C150Y and N125Y.

Below are some of the main recommendation for further improvement on the APC performance as well as plant stability and reliability:

1. To automate 12TIC1113 to ensure better temperature control and stability at back end cooler, and hence improving suction temperature.
2. To implement automation on dewax procedure at back end cooler as it can reduce process nonlinearity to enable item 1.
3. To standardize controller algorithm on DCS to be at 1 fix algorithm instead of Auto Determination
 2. Auto Determination 2 will use I-PD when it is in AUTO and RCAS. However, it will change to PI-D when controller mode is changed to CAS.

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APPENDIX A

Controller Tuning – As per December 2020

Current APC		New APC Commissioning				
Recipe	Parameter	Current Aim	cv_lower_operator	cv_upper_operator	cv_upper_engin	cv_lower_engin
C150Y_Target	MFI	5	4.3	5.8	NULL	NULL
	GLOSS	NULL	62	72	NULL	NULL
	PRODSPEX	NULL	32	36	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
N125Y_Target	MFI	2.5	2.2	2.9	NULL	NULL
	GLOSS	NULL	52	62	NULL	NULL
	PRODSPEX	NULL	32	36	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
N103X_Target	MFI	0.29	0.25	0.33	NULL	NULL
	GLOSS	NULL	34	44	NULL	NULL
	PRODSPEX	NULL	29	30	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
N109X_Target	MFI	0.85	0.75	0.98	NULL	NULL
	GLOSS	NULL	48	58	NULL	NULL
	PRODSPEX	NULL	30	32	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
M022X_Target	MFI	22	19	25	NULL	NULL
	GLOSS	NULL	0	100	NULL	NULL
	PRODSPEX	32	30	33	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
M050X_Target	MFI	48	44	57	NULL	NULL
	GLOSS	NULL	0	100	NULL	NULL
	PRODSPEX	NULL	31	33	NULL	NULL
	DIEPRESS	NULL			NULL	NULL
C170X_3.5_Target	MFI	3.5	3.2	3.8	NULL	NULL
	GLOSS	NULL	0	100	NULL	NULL
	PRODSPEX	35.5	32	36	NULL	NULL
	DIEPRESS	NULL			NULL	NULL

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Current APC		MV's					
Recipe		Parameter	Current Aim	mv_lower_oper_limit	mv_upper_oper_limit	mv_lower_eng_lim	mv_upper_eng_lim
C150Y_Target		APCTA	NULL	500	600	100	1500
		APTEMP1	NULL	297	299	270	307
		APTEMP2	NULL	297	299	270	307
		APTEMP3	NULL	297	299	270	307
		APTEMP4	NULL	295	298	265	307
		APPURGE	NULL	600	1200	400	1400
N125Y_Target		APCTA	NULL	420	470	100	1500
		APTEMP1	NULL	297	298	270	307
		APTEMP2	NULL	297	298	270	307
		APTEMP3	NULL	297	298	270	307
		APTEMP4	NULL	295	298	265	307
		APPURGE	NULL	600	1200	400	1400
N103X_Target		APCTA	NULL	215	260	100	1500
		APTEMP1	NULL	283	287	270	307
		APTEMP2	NULL	287	293	270	307
		APTEMP3	NULL	287	293	270	307
		APTEMP4	NULL	287	293	265	307
		APPURGE	NULL	600	1200	400	1400
N109X_Target		APCTA	NULL	330	370	100	1500
		APTEMP1	NULL	287	291	270	307
		APTEMP2	NULL	287	291	270	307
		APTEMP3	NULL	287	291	270	307
		APTEMP4	NULL	287	291	265	307
		APPURGE	NULL	600	1200	400	1400
M022X_Target		APCTA	NULL	850	1000	100	1500
		APTEMP1	NULL	296	298	270	307
		APTEMP2	NULL	296	298	270	307
		APTEMP3	NULL	296	298	270	307
		APTEMP4	NULL	270	278	265	307
		APPURGE	NULL	600	1200	400	1400
M050X_Target		APCTA	NULL	1030	1100	100	1500
		APTEMP1	NULL	296	298	270	307
		APTEMP2	NULL	296	298	270	307
		APTEMP3	NULL	296	298	270	307
		APTEMP4	NULL	270	278	265	307
		APPURGE	NULL	600	1200	400	1400
C170X_3.5_Target		APCTA	NULL	460	490	100	1500
		APTEMP1	NULL	298	300	270	307
		APTEMP2	NULL	298	300	270	307
		APTEMP3	NULL	298	300	270	307
		APTEMP4	NULL	298	300	265	307
		APPURGE	NULL	600	1200	400	1400

 PETRONAS	GROUP TECHNICAL SOLUTIONS	
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APC Functional Design Specifications (FDS)

		New APC Commissioning												
Recipe	Parameter	cv_time_constant	cv_noise_ratio	cv_typical_disturbance	cv_ideal_restring_value	cv_out_of_service_valu	cv_out_of_service_swit	cv_setpoint_equal_con	cv_ss_tuning_method	cv_limit_equal_concern	cv_trajectory_factor	cv_irv_equal_concern	cv_operator_request	cv_calculated_status
C150Y_Tuning	MFI	1	0.01	0.01	3	5	0	0.05	3	0.01	0	0.01	1	1
	GLOSS	1	0.5	0.5	0	66	0	10	0	5	0.5	0	1	1
	PRODSPEX	1	0.5	0.5	2	34.5	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.01	0.01	3	48.54	0	0.025	3	0.005	0	0.943	1	1
N125Y_Tuning	MFI	1	0.005	0.005	3	2.5	0	0.025	3	0.005	0	0.005	1	1
	GLOSS	1	0.5	0.5	0	56	0	10	0	5	0.5	0	1	1
	PRODSPEX	1	0.5	0.5	2	34.03	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.005	0.05	3	54.06	0	0.0125	3	0.005	0	0.868	1	1
N103X_Tuning	MFI	1	0.001	0.001	3	5	0	0.002	3	0.001	0	0.001	1	1
	GLOSS	1	0.5	0.5	0	65	0	10	0	5	0.5	0	1	1
	PRODSPEX	1	0.5	0.5	2	33	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.001	0.001	3	67.61	0	0.001	3	0.005	0	0.733	1	1
N109X_Tuning	MFI	1	0.003	0.003	3	0.85	0	0.001	3	0.001	0	0.002	1	1
	GLOSS	1	0.5	0.5	0	52	0	10	0	5	0.5	0	1	1
	PRODSPEX	1	0.5	0.5	2	30	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.003	0.003	3	60.62	0	0.0005	3	0.005	0	0.565	1	1
M022X_Tuning	MFI	1	0.05	0.05	3	22	0	0.5	3	0.05	0	0.05	1	1
	GLOSS	1	0.5	0.5	0	65	0	10	0	5	0.5	0	0	0
	PRODSPEX	1	0.5	0.5	2	33	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.05	0.05	3	38.78	0	0.25	3	0.005	0	1.229	1	1
M050X_Tuning	MFI	1	0.1	0.1	3	50	0	0.5	3	0.1	0	0.1	1	1
	GLOSS	1	0.5	0.5	0	65	0	10	0	5	0.5	0	0	0
	PRODSPEX	1	0.5	0.5	2	33	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.1	0.1	3	28.23	0	0.25	3	0.005	0	0.493	1	1
C170X_3.5_Tuning	MFI	1	0.005	0.005	3	3.5	0	0.025	3	0.005	0	0.005	1	1
	GLOSS	1	0.5	0.5	0	56	0	10	0	5	0.5	0	0	0
	PRODSPEX	1	0.5	0.5	2	33.5	0	10	0	5	0.5	0	1	1
	DIEPRESS	1	0.005	0.005	3	51.95	0	0.0125	3	0.005	0	0.596	1	1

Current APC	New APC Commissioning																
Recipe	Parameter	mv_steady_state_step_limit	mv_move_resolution	mv_lower_step_limit	mv_manual_mode	mv_move_suppression_factor	mv_upper_step_limit	mv_out_of_service_switch	mv_loop_status	mv_irv_equal_concern_error	mv_setpoint_equal_concern_error	mv_ideal_restring_value_type	mv_ss_tuning_method	mv_out_of_service_value	mv_calculated_status	mv_anti_windup_code	mv_trajectory_factor
		MSTEP	MVRES	MXMDW	MAN	MVSUP	MXMUP	MOSSW	MLPST	MIREC	MDVEC	MIRTP	MSSTP	MSRVL	MSTAT	MAWS	MTRF
C150Y_Tuning	APCTA	1500	0	10	0	1	10	0	1	0	20	0	0	530	0	0	0
	APTEMP1	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP2	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP3	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP4	310	0	0.5	0	5	0.5	0	1	0	3	0	0	296	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.31433333	0	20	0.31433333	0	1	0	0.31433333	0	0	48.54	0	0	0
N125Y_Tuning	APCTA	1500	0	10	0	0.025	10	0	1	0	20	0	0	430	0	0	0
	APTEMP1	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP2	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP3	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP4	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	296	0	0	0
	APPURGE	1500	0	25	0	2.5	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.28933333	0	2.5	0.28933333	0	1	0	0.28933333	0	0	54.06	0	0	0
N103X_Tuning	APCTA	1500	0	5	0	0.025	5	0	1	0	10	0	0	235	0	0	0
	APTEMP1	310	0	0.5	0	2.5	0.5	0	1	0	3	0	0	285	0	0	0
	APTEMP2	310	0	0.5	0	2.5	0.5	0	1	0	3	0	0	290	0	0	0
	APTEMP3	310	0	0.5	0	2.5	0.5	0	1	0	3	0	0	290	0	0	0
	APTEMP4	310	0	0.5	0	2.5	0.5	0	1	0	3	0	0	290	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.24433333	0	5	0.24433333	0	1	0	0.24433333	0	0	67.61	0	0	0
N109X_Tuning	APCTA	1500	0	5	0	0.025	5	0	1	0	5	0	0	355	0	0	0
	APTEMP1	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	289	0	0	0
	APTEMP2	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	289	0	0	0
	APTEMP3	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	289	0	0	0
	APTEMP4	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	289	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.18833333	0	5	0.18833333	0	1	0	0.18833333	0	0	60.62	0	0	0
M022X_Tuning	APCTA	1500	0	15	0	0.025	15	0	1	0	25	0	0	840	0	0	0
	APTEMP1	310	0	0.1	0	5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP2	310	0	0.1	0	5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP3	310	0	0.1	0	5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP4	310	0	0.1	0	5	0.1	0	1	0	3	0	0	275	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.40966667	0	2.5	0.40966667	0	1	0	0.40966667	0	0	38.78	0	0	0
M050X_Tuning	APCTA	1500	0	15	0	0.025	15	0	1	0	30	0	0	1015	0	0	0
	APTEMP1	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP2	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP3	310	0	0.5	0	5	0.5	0	1	0	3	0	0	298	0	0	0
	APTEMP4	310	0	1	0	5	0.5	0	1	0	3	0	0	275	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.16433333	0	2.5	0.16433333	0	1	0	0.16433333	0	0	28.23	0	0	0
C170X_3.5_Tuning	APCTA	1500	0	5	0	0.025	5	0	1	0	20	0	0	460	0	0	0
	APTEMP1	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP2	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP3	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	298	0	0	0
	APTEMP4	310	0	0.1	0	2.5	0.1	0	1	0	3	0	0	296	0	0	0
	APPURGE	1500	0	25	0	1	25	0	1	0	50	1	0	1100	0	0	0
	DIEPRESS	80	0	0.19866667	0	2.5	0.19866667	0	1	0	0.19866667	0	0	51.95	0	0	0